

Remaster Ubuntu

to Get a Customised Distribution



In this article, we look at how to create a custom GNU/Linux distribution based on Ubuntu.

Customisability is a feature most people find amazing about Free Software. The software can be tweaked to any extent to suit individual tastes. Though this has created confusion with newbies, it has always helped the software to grow. We can create modified versions suitable for a specific purpose, eliminating the need to re-invent the wheel—i.e., creating parts that are already available.

Why not apply this to a distro? You may be a sysadmin who needs to install a specific set of tools across all company systems, a student who needs an elegant solution to distribute Linux-based software, or a geek who loves to tweak software just for the fun of it.

I first thought of a custom distro when some of my classmates complained about the difficulty in obtaining and installing software tools needed for coursework in GNU/Linux. Most of them were new users of the GNU/Linux platform, so they wanted a simple solution that suited everyone. My friends and I, while looking for solutions, thought of looking for a distribution in which all the software we needed was pre-installed. However, a few Google searches disappointed us. There are distros such as Uberstudent, which are designed for students, but there were none that were specifically for technical students. That's when we decided to roll out our own, and we're happy that we did.

Customisation of the base distro

We selected Ubuntu as our base system, for its manageability, and also because most of our users are new to GNU/Linux. To re-build the distribution we used Remastersys, a useful software that packs your existing system into a 'live' disk image. Let's look at how we did this.

To create a custom GNU/Linux distro, you need to install and customise the base distro first; so we installed a fresh copy of Ubuntu onto a system that was going to be our 'build' machine. Next, we applied all our customisation to the system, installed the software we needed, and applied customisations to the desktop.

With that done, it was time to install Remastersys. To do this, add the repository to your apt sources by adding the following line to the file `/etc/apt/sources.list`, as you can see in Figure 1:

```
deb http://www.geekconnection.org/remastersys/repository karmic/
```

(This line differs based on the distro version; see <http://www.geekconnection.org/remastersys/ubuntu.html>.)

After saving `sources.list`, execute the following code in a terminal to reload the package lists and install Remastersys:

```
sudo apt-get update && sudo apt-get install remastersys
```

Preserving customised user settings

When building the live image, by default, Remastersys excludes the user's home folders, so as to not include personal settings/data into the build. This will cause any interface (desktop) customisation in your user account to be reset to Ubuntu defaults when the live image is booted. How do we maintain these customisations in the live image as well? There is a standard 'skeleton' folder, `/etc/skel`, that is copied as the home folder for each new user created on a GNU/Linux system (see Figure 2). In our user account's home folder, interface customisations are stored in the (hidden) `.gnome2` and `.gconf` folders. For example, the wallpaper setting is